

MOHAMMED MAKSOOD ALAM M

AI Engineer

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Summary

AI Engineer building production Generative AI (GenAI) systems. Independently designed and shipped a RAG chatbot serving 500+ users, a hybrid-retrieval question generation engine, and a multilingual document ingestion pipeline, using Python, Django, LangChain and Qdrant with OpenAI, Gemini and Anthropic models.

Skills

- **AI & LLM** – Large Language Models (LLMs), Applied Machine Learning, NLP, RAG Pipelines, Hybrid Retrieval (Keyword + Semantic Search), LLM Orchestration (LangChain, LangGraph), Agentic Frameworks (Google ADK, CrewAI), Multi-Agent Systems, Prompt Engineering, LLM Observability (LangSmith), OpenAI / Gemini / Anthropic APIs, Ollama, Sentence Transformers, IndicBERT Embeddings
- **Languages** – Python, JavaScript
- **Backend & Infra** – Django, REST APIs, Node.js, Express.js, Celery, Redis, Kafka/Redpanda, Socket.IO, Docker, RunPod, AWS S3/CloudFront
- **Databases** – Vector Databases (Qdrant, ChromaDB), Neo4j (graph), MongoDB, MySQL
- **Tools** – Git/GitHub, Gradio, OpenCV, PyMuPDF, Tesseract, Postman, Claude Code, Cursor

Experience

AI Engineer | EdgeUp | May 2025 – Present

AI Tutoring Chatbot (RAG)

Tech: LangChain, Socket.IO, Kafka/Redpanda, Qdrant, Redis, Celery, OpenAI GPT-4o-mini, Gemini 2.5 Flash

- Built a curriculum-grounded RAG chatbot over course books and past question papers, serving 500+ students
- Cut response time from 10s to 4–7s by switching from full-response delivery to token-by-token streaming, so students see the answer forming immediately
- Architected a real-time messaging layer (Socket.IO + Kafka/Redpanda + Redis), load-tested with the QA team via JMeter at 1,000 concurrent users
- Added automatic fallback from GPT-4o-mini to Gemini 2.5 Flash on provider errors and timeouts, keeping the chatbot available during OpenAI outages

Intelligent Question Generation Engine (Hybrid RAG)

Tech: Gemini 2.5 Pro, Qdrant, Neo4j, Redis, Sentence Transformers, IndicBERT

- Built a hybrid-retrieval (keyword + semantic search) RAG engine generating thousands of MCQs across 22 courses and competitive exams (UPSC, TNPSC, SSC, Banking, RRB)
- Raised expert-reviewer acceptance of generated MCQs from 90% to near-100% by refining prompts across the multi-agent generation and validation pipeline, validated by an in-house research team of UPSC/TNPSC aspirants
- Cut LLM API spend by 30%+ (verified on API bills) by caching repeated requests in Redis, and added rate limiting to stay within provider quotas

Automated Exam Grading System

Tech: OpenCV, Gemini, Celery Beat, AWS S3/CloudFront

- Built the OpenCV OMR module that reads bubble-sheet answers from scanned exam scripts, feeding a team-built auto-grading pipeline (OMR + Gemini) that cut grading time from hours to minutes

Multi-Course PDF Knowledge Ingestion Pipeline

Tech: Gemini 2.5 Flash, Qdrant, Neo4j, PyMuPDF, Tesseract OCR, Gradio, IndicBERT

- Designed a PDF ingestion pipeline for 500+ documents across 13+ subjects, indexing chunks in Qdrant with multilingual IndicBERT embeddings and building a Neo4j knowledge graph
- Added a Tesseract OCR fallback for scanned PDFs where PyMuPDF text extraction fails
- Built a Gradio console for on-demand MCQ generation via Gemini 2.5 Flash

Junior Developer (Intern) | Zaryah Angels Pvt. Ltd. | Jan 2025 – Mar 2025

- Designed system prompts for the question-generation MVP
- Compared vector databases (Weaviate vs Qdrant) for the MVP's retrieval layer; Qdrant was adopted and later used in production

Education

- Bachelor of Computer Applications (BCA) – The New College, Chennai | 2025 | GPA: 7.2/10
- HSC – Justice Basheer Ahmed Sayeed Memorial Matric. Hr. Sec. School | 2022 | 83.5%

Certifications: Android App Development – Techobytes / IIT Madras · MongoDB Atlas – MongoDB · SQL & Relational Databases – IBM